



Ecole Supérieure en Informatique

ESI-Sidi Bel Abbès

Lab Report 06

Network Traffic Analysis with Wireshark & tcpdump

*Computer Systems and Network Security
Academic Year 2025–2026*

Student:

Manel Mostefaoui

Instructor:

Mohammed Salih Bendella

1. Objectives :

Learn how to Secure SSH remote access to prevent unauthorized access while maintaining legitimate user access

2. Part 1 – Installing and Enabling SSH

2.1 Procedure

```
dpkg -l | grep openssh
sudo apt update
sudo apt install -y openssh-server
sudo systemctl enable ssh
sudo systemctl start ssh
systemctl status ssh
ssh -v kali@192.168.24.128
```

2.2 Answers

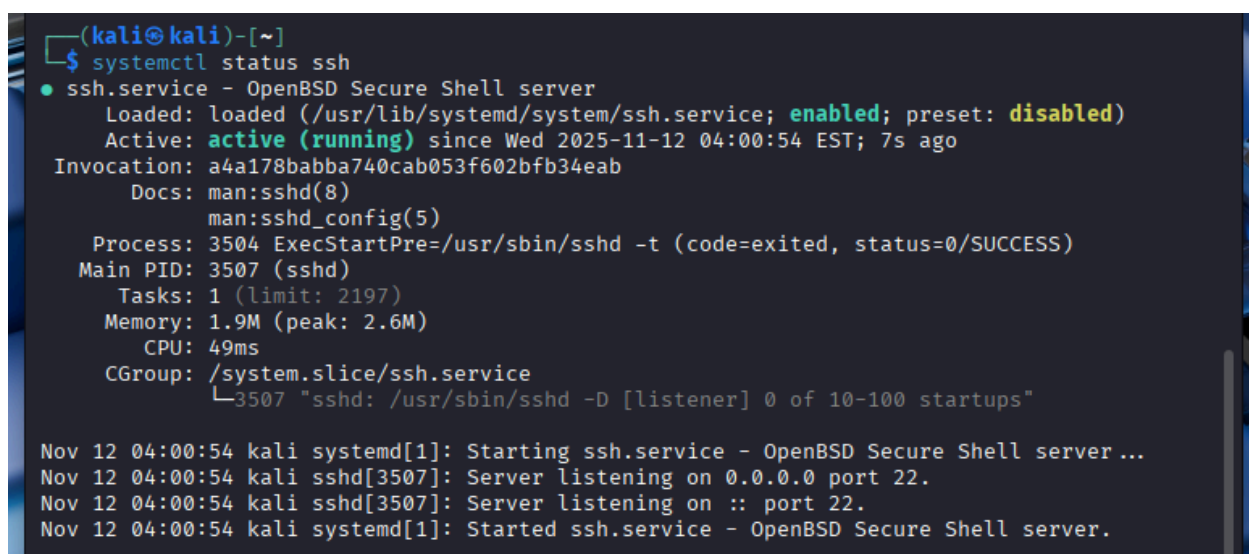
Question 1: What port is SSH listening on by default?

The port SSH listening on by default is port 22.

Question 2: What protocol version does SSH use by default (v1 or v2)?

The protocol version is v2 because (debug1: Remote protocol version 2.0, remote software version OpenSSH_10.0p2 Debian-8)

2.3 Screenshoot :



```
(kali㉿kali)-[~]
$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-11-12 04:00:54 EST; 7s ago
  Invocation: a4a178babba740cab053f602bfb34eab
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 3504 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 3507 (sshd)
    Tasks: 1 (limit: 2197)
   Memory: 1.9M (peak: 2.6M)
      CPU: 49ms
   CGroup: /system.slice/ssh.service
           └─3507 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

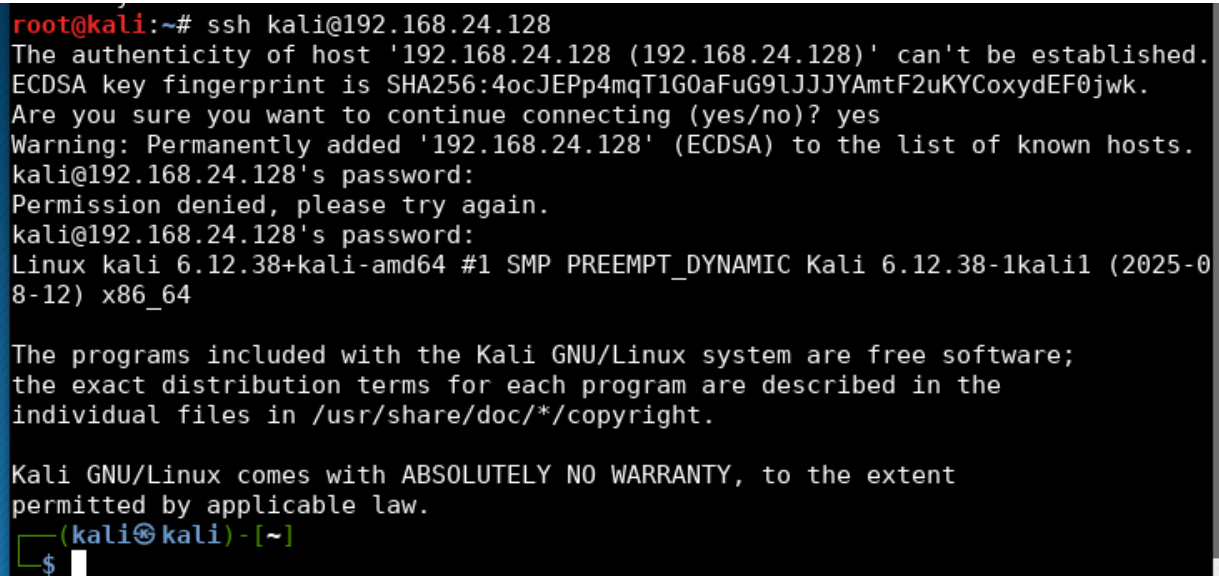
Nov 12 04:00:54 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Nov 12 04:00:54 kali sshd[3507]: Server listening on 0.0.0.0 port 22.
Nov 12 04:00:54 kali sshd[3507]: Server listening on :: port 22.
Nov 12 04:00:54 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

3. Part 2 – Basic Connection Test

2.1 Procedure

```
ssh kali@192.168.24.128
```

2.2 Screenshot :



```
root@kali:~# ssh kali@192.168.24.128
The authenticity of host '192.168.24.128 (192.168.24.128)' can't be established.
ECDSA key fingerprint is SHA256:4ocJEpp4mqTIG0aFuG9lJJYAmtF2uKYCoxydEF0jwk.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.24.128' (ECDSA) to the list of known hosts.
kali@192.168.24.128's password:
Permission denied, please try again.
kali@192.168.24.128's password:
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
(kali@kali) - [~]
$
```

4. Part 3 – Securing SSH Configuration

4.1 Procedure

```
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak.$(date +%F_%T)
sudo nano /etc/ssh/sshd_config
```

4.2 Answers

Challenge 1: Modify the configuration file to allow access using the root user from the Metasploitable machine on the default port.

Solution : Add line: PermitRootLogin yes and Port 22

```
kali@kali: ~
Session Actions Edit View Help
GNU nano 8.6 /etc/ssh/sshd_config *

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C L
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^D Justify    ^_ G
```

Challenge 2: Use port 666 for the SSH connection.

Solution : Add line: Port 666

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 666
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

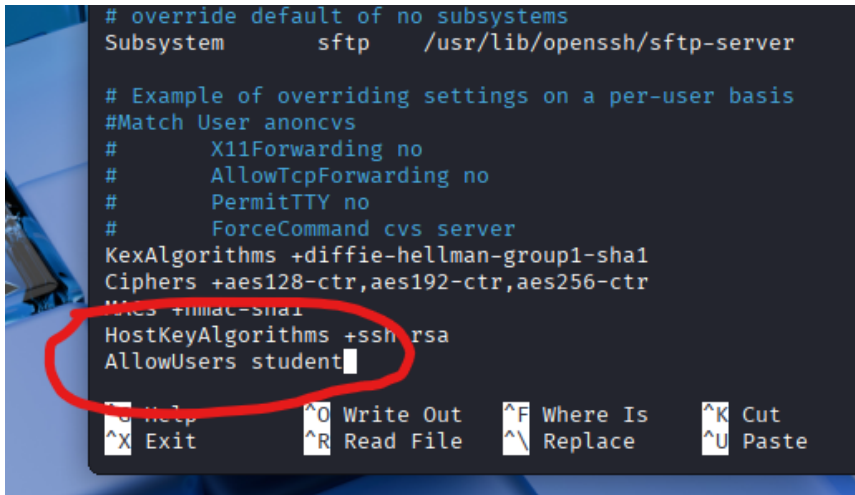
# Logging
```

Challenge 3: Create a “student” user on Kali and add them to the list of users authorized for SSH connection. Then, specify the listening address to be that of the Metasploitable machine.

Solution :

sudo adduser student

Add line: AllowUsers student

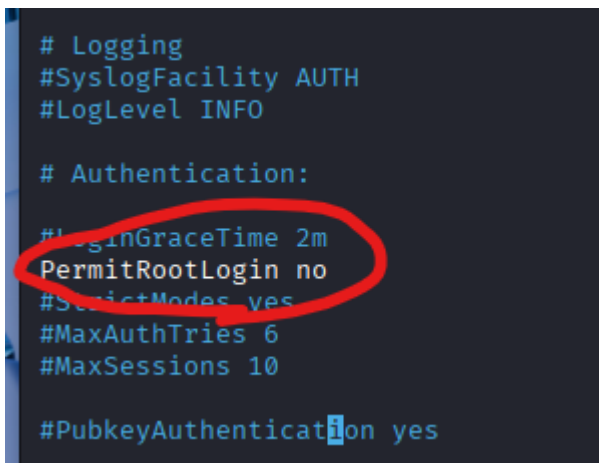
A terminal window showing the editing of the /etc/ssh/sshd_config file. The user has added 'AllowUsers student' at the end of the file. The line is circled in red. The terminal also shows various SSH configuration options like 'Subsystem sftp', 'X11Forwarding no', 'AllowTcpForwarding no', 'PermitTTY no', 'ForceCommand cvs server', 'KexAlgorithms', 'Ciphers', 'MACs', and 'HostKeyAlgorithms'.

```
# override default of no subsystems
Subsystem        sftp        /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server
KexAlgorithms +diffie-hellman-group1-sha1
Ciphers +aes128-ctr,aes192-ctr,aes256-ctr
MACs +hmac-sha1
HostKeyAlgorithms +ssh-rsa
AllowUsers student
```

Challenge 4: Block SSH connections from the root user.

Solution : Modify line : PermitRootLogin no

A terminal window showing the editing of the /etc/ssh/sshd_config file. The user has modified the 'PermitRootLogin' setting to 'no'. The line is circled in red. The terminal also shows various SSH configuration options like 'Logging', 'SyslogFacility AUTH', 'LogLevel INFO', 'Authentication:', 'LoginGraceTime 2m', 'StrictModes yes', 'MaxAuthTries 6', 'MaxSessions 10', and 'PubkeyAuthentication yes'.

```
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:
#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

Challenge 5: Add the following parameters:

Forces SSH v2 (more secure).

Blocks empty passwords.

Limits login attempts.

Disconnects after 30s if no login.

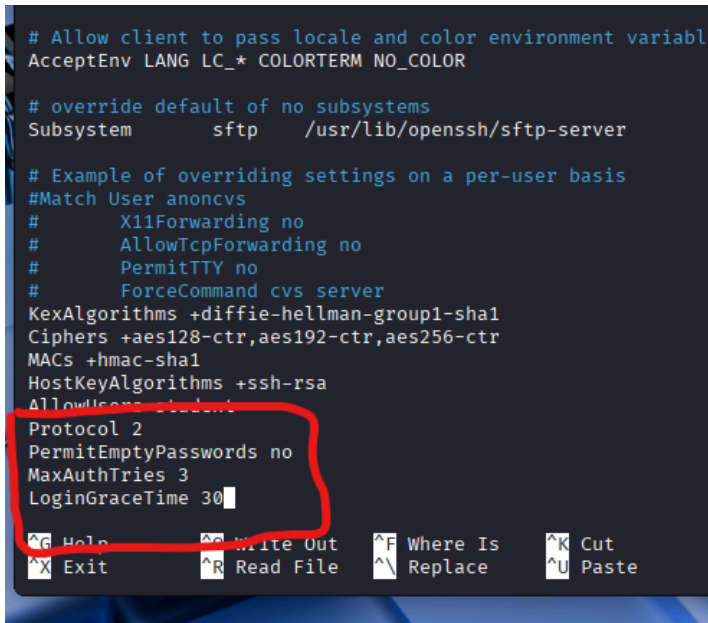
solution : add lines :

Protocol 2

PermitEmptyPasswords no

MaxAuthTries 3

LoginGraceTime 30



```
# Allow client to pass locale and color environment variables
AcceptEnv LANG LC_* COLORTERM NO_COLOR

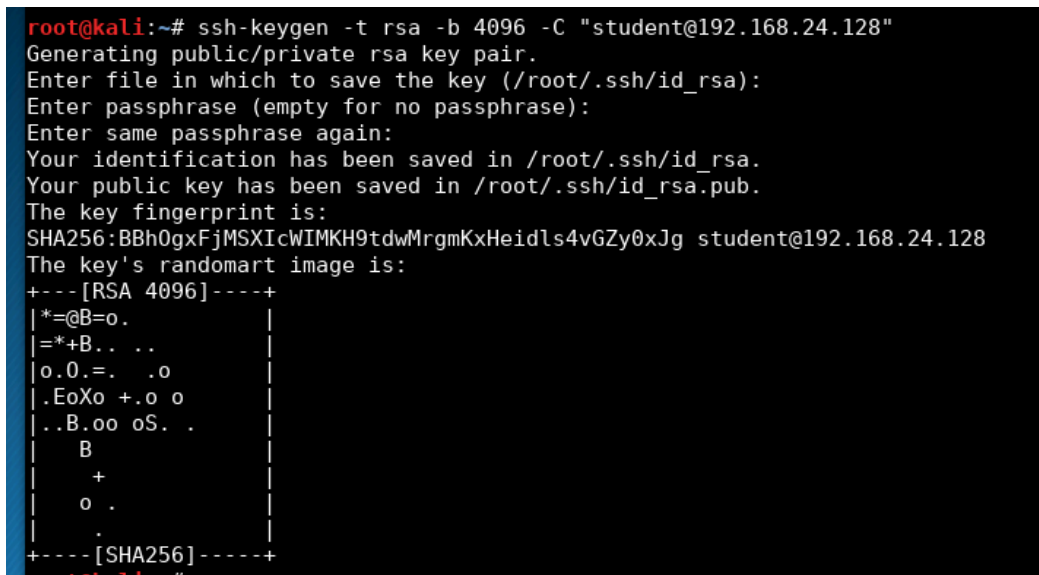
# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
KexAlgorithms +diffie-hellman-group1-sha1
Ciphers +aes128-ctr,aes192-ctr,aes256-ctr
MACs +hmac-sha1
HostKeyAlgorithms +ssh-rsa
AllowDups no
Protocol 2
PermitEmptyPasswords no
MaxAuthTries 3
LoginGraceTime 30
```

5. Part 4 – Using SSH Key Authentication

5.1 Generate a key pair (on the client)

```
ssh-keygen -t rsa -b 4096 -C "student@192.168.24.128"
```



```
root@kali:~# ssh-keygen -t rsa -b 4096 -C "student@192.168.24.128"
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa.
Your public key has been saved in /root/.ssh/id_rsa.pub.
The key fingerprint is:
SHA256:BBh0gxFjMSXicWIMKH9tdwMrgmKxHeidls4vGZy0xJg student@192.168.24.128
The key's randomart image is:
+---[RSA 4096]-----+
|*=@B=o.             |
|*=+B.. ..          |
|o.O.=. .o          |
|.EoXo +.o o        |
|..B.oo oS. .       |
|  B                 |
|  +                 |
|  o .               |
|  .                 |
+---[SHA256]-----+
root@kali:~#
```

5.2 Copy the public key to the server

```
(kali@kali)-[~]
$ ssh-copy-id -p 666 student@192.168.24.128

/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kali/.ssh/id_rsa.pub"
The authenticity of host '[192.168.24.128]:666 ([192.168.24.128]:666)' can't be established.
ED25519 key fingerprint is SHA256:187mKnFvq2hCufo4mtc992Rp+D3Ltg9w3Vcl1HfhAo8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that
are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it
is to install the new keys
student@192.168.24.128's password:
Permission denied, please try again.
student@192.168.24.128's password:
Permission denied, please try again.
student@192.168.24.128's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -p 666 'student@192.168.24.128'"
and check to make sure that only the key(s) you wanted were added.
```

5.3 Answers

Challenge: Disable password login completely, enables authentication using public key pairs on sshd_config and try to connect without password.

Solution : Modify these lines :

PubkeyAuthentication yes

PasswordAuthentication no

```
PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

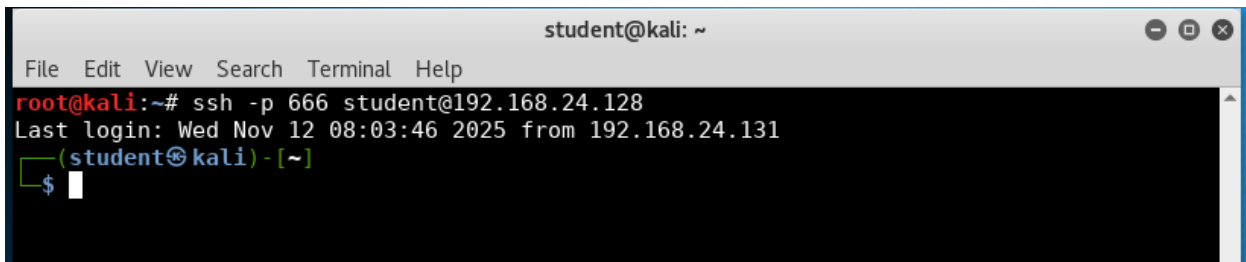
#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication no
#PermitEmptyPasswords no
```

5.4 Result :



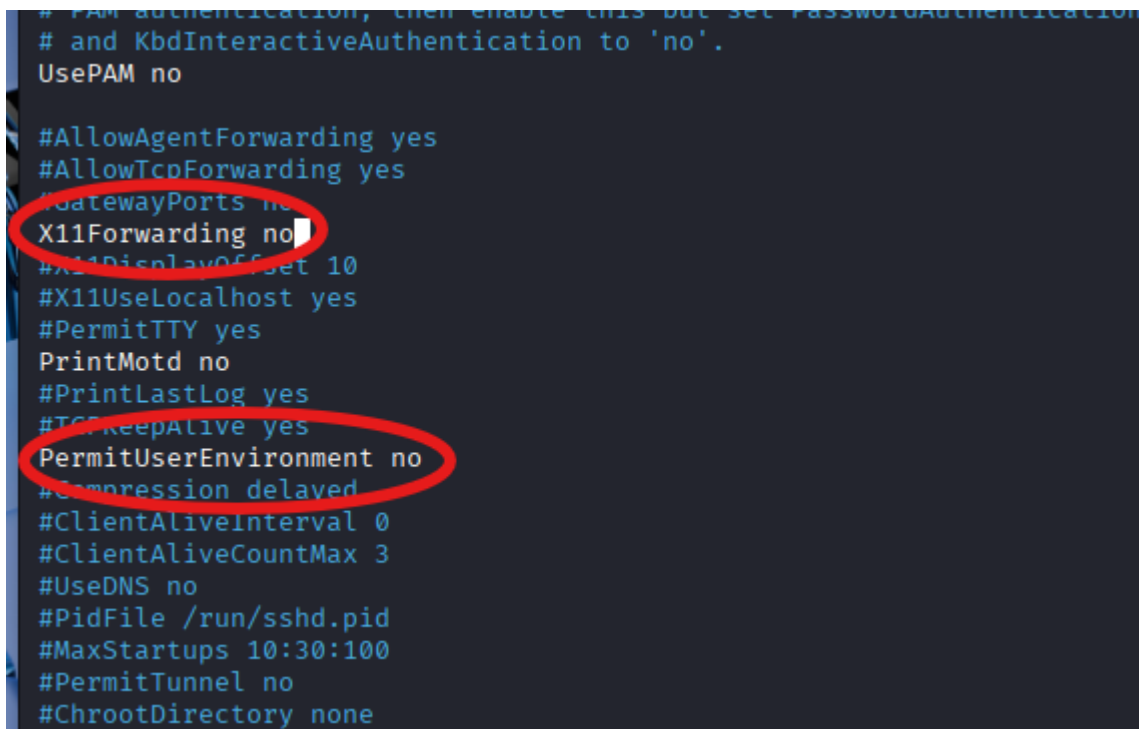
```
student@kali: ~
File Edit View Search Terminal Help
root@kali:~# ssh -p 666 student@192.168.24.128
Last login: Wed Nov 12 08:03:46 2025 from 192.168.24.131
(student@kali) - [~]
$
```

Question: Why is key-based authentication more secure than password-based authentication?

Answer : key-based authentication more secure than password-based authentication because it uses cryptographically strong keys that are resistant to brute force attacks and never transmit secrets over the network.

6. Part 5 – Additional Hardening

5.1 Disable unused features:



```
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM no

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding no
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
```


5.2 Restrict SSH access using firewall (UFW)

```
sudo apt install -y ufw
```

```
sudo ufw allow 666/tcp (allow TCP traffic on port 666 in the UFW.)
```

```
sudo ufw enable
```

```
sudo ufw status
```

```
(kali㉿kali)-[~]
$ sudo ufw allow 666/tcp
Rules updated
Rules updated (v6)

(kali㉿kali)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(kali㉿kali)-[~]
$ sudo ufw status
Status: active

To          Action      From
--          -
666/tcp     ALLOW      Anywhere
666/tcp (v6) ALLOW      Anywhere (v6)
```

5.3 Enable fail2ban (to block brute-force attacks)

```
(kali㉿kali)-[~]
$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/sy
stemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib
/systemd/system/fail2ban.service'.

(kali㉿kali)-[~]
$ sudo cat /etc/fail2ban/jail.conf | grep ssh
# ssh-iptables jail the following (uncommented) would appear in the .local file.
# [sshd]
[sshd]
# To use more aggressive sshd modes set filter parameter "mode" in jail.local:
# See "tests/files/logs/sshd" or "filter.d/sshd.conf" for usage example and details.
port      = ssh
logpath   = %(sshd_log)s
backend   = %(sshd_backend)s
port      = ssh
[selinux-ssh]
port      = ssh
port      = ssh,http,https
```

7. Part 6 – Monitoring and Logs

```
(kali㉿kali)-[~]  
$ sudo journalctl -u ssh.service --no-pager  
Nov 12 04:00:54 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...  
Nov 12 04:00:54 kali sshd[3507]: Server listening on 0.0.0.0 port 22.  
Nov 12 04:00:54 kali sshd[3507]: Server listening on :: port 22.  
Nov 12 04:00:54 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
Nov 12 04:07:21 kali sshd-session[6369]: Connection closed by 192.168.24.128 port 54296 [preauth]  
Nov 12 04:15:09 kali sshd-session[9851]: Unable to negotiate with 192.168.24.129 port 54403: no matching key exchange method found. Their offer: diffie-hellman-group-exchange-sha256,diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1 [preauth]  
Nov 12 04:15:12 kali sshd-session[9877]: Unable to negotiate with 192.168.24.129 port 54404: no matching key exchange method found. Their offer: diffie-hellman-group-exchange-sha256,diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1 [preauth]  
Nov 12 04:17:31 kali sshd-session[10920]: Unable to negotiate with 192.168.24.129 port 54405: no matching key exchange method found. Their offer: diffie-hellman-group-exchange-sha256,diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1 [preauth]  
Nov 12 04:25:58 kali sshd-session[14485]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.24.130 user=kali  
Nov 12 04:25:58 kali sshd-session[14485]: pam winbind(sshd:auth): getting password (0x0000
```

8. Part 7 – Optional Challenge

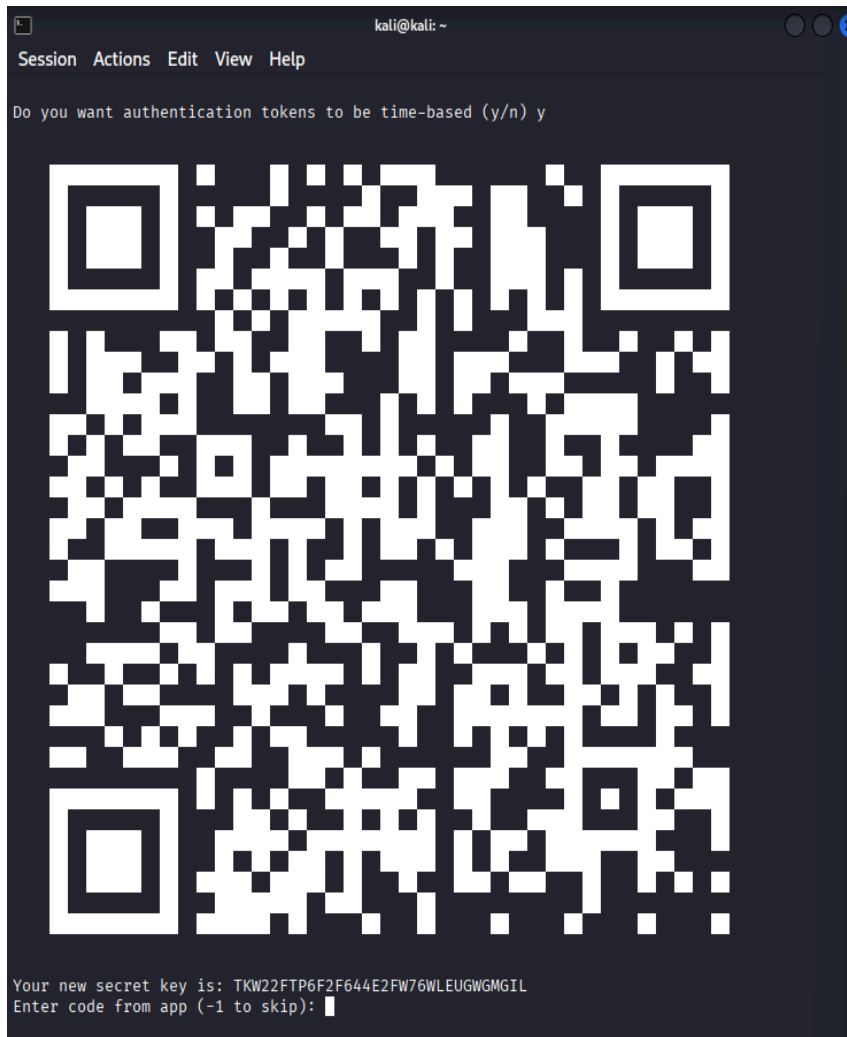
Step 1: Install Google Authenticator on the Server

```
sudo apt-get install libpam-google-authenticator
```

```
(kali㉿kali)-[~]  
$ su - student  
Password:  
(student㉿kali)-[~]
```

Step 2: Configure Google Authenticator for a User

```
google-authenticator
```



```
Your new secret key is: TKW22FTP6F2F644E2FW76WLEUGWGMGIL
Enter code from app (-1 to skip): 954821
Code confirmed
Your emergency scratch codes are:
14616801
77680016
56944607
31404271
61370704

Do you want me to update your "/home/kali/.google_authenticator" file? (y/n) y

Do you want to disallow multiple uses of the same authentication
token? This restricts you to one login about every 30s, but it increases
your chances to notice or even prevent man-in-the-middle attacks (y/n) y

By default, a new token is generated every 30 seconds by the mobile app.
In order to compensate for possible time-skew between the client and the server,
we allow an extra token before and after the current time. This allows for a
time skew of up to 30 seconds between authentication server and client. If you
experience problems with poor time synchronization, you can increase the window
from its default size of 3 permitted codes (one previous code, the current
code, the next code) to 17 permitted codes (the 8 previous codes, the current
code, and the 8 next codes). This will permit for a time skew of up to 4 minutes
between client and server.
Do you want to do so? (y/n) y

If the computer that you are logging into isn't hardened against brute-force
login attempts, you can enable rate-limiting for the authentication module.
By default, this limits attackers to no more than 3 login attempts every 30s.
Do you want to enable rate-limiting? (y/n) y
```

Step 3: Configure SSH to Use Google Authenticator

```
sudo nano /etc/pam.d/sshd
```

Add the following line to the top of the file to enable Google Authenticator for SSH

```
auth required pam_google_authenticator.so nullok
auth required pam_permit.so
```

Step 4: Configure SSH to Allow 2FA

Open the SSH server configuration file:

```
sudo nano /etc/ssh/sshd_config
```

Add or modify the following line to allow PAM authentication :

Step 5: Restart SSH Service

```
sudo systemctl restart ssh
```

9. The results

```
(kali㉿kali)-[~]  
$ ssh -p 666 student@192.168.20.197  
(student@192.168.20.197) Verification code:  
(student@192.168.20.197) Password:  
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64  
  
The programs included with the Kali GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Wed Nov 12 10:46:20 2025 from 192.168.20.197  
(student㉿kali)-[~]  
$
```

